

MAT1002 Lecture 18,

Outline

- Results related to second partials (continued)
- Taylor's theorem
- Double integrals on rectangles
- Double integrals on general bounded region

The Second Derivative Test

Theorem (Second Derivative Test)

Let f be a function whose second partial derivatives are all continuous on an open ^{disk} ball centered at (a, b) . Suppose that $\nabla f(a, b) = \vec{0}$ (so (a, b) is a critical point of f). Let

$$H := H(a, b) := f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2.$$

- ▶ If $H > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b) .
- ▶ If $H > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b) .
- ▶ If $H < 0$, then f has no local extremum at (a, b) ; that is, (a, b) is a saddle point of f .

Proof: • Consider $h := \Delta x$ and $k := \Delta y$ as "small changes".

• From last time, we know that

$$f(a+h, b+k) = f(a, b) + f_x(a, b)h + f_y(a, b)k + \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2) \Big|_{(x,y)=(a+ch, b+ck)}$$

for some $c \in (0, 1)$.

• Since $\nabla f(a, b) = (0, 0)$,

$$E(a, b) = f(a+h, b+k) - f(a, b) = \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2) \Big|_{(x,y)=(a+ch, b+ck)} \quad \textcircled{1} \geq 0$$

$Q(c) = Q_{h,k}(c)$

• Let $Q(t) := Q_{h,k}(t) := \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2) \Big|_{(x,y)=(a+th, b+tk)}$.

$$Q(0) =$$

- Suppose $Q(0) > 0$. Since f_{xx} , f_{yy} and f_{xy} are all cts at (a,b) , for all (h,k) sufficiently close to $(0,0)$, $Q(c) > 0$; in this case, by ①, $f(a,b)$ is a local minimum.
- Similarly, if $Q(0) < 0$, then $f(a,b)$ is a local maximum.
- Hence, we investigate $Q(0)$.

$$Q(0) = \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2)|_{(x,y)=(a,b)}$$

$$\Rightarrow f_{xx}(a,b)Q(0) = \frac{1}{2}(f_{xx}^2h^2 + 2f_{xx}f_{xy}hk + f_{xx}f_{yy}k^2)|_{(x,y)=(a,b)}$$

$$= \frac{1}{2}[(f_{xx}h + f_{xy}k)^2 + (f_{xx}f_{yy} - f_{xy}^2)k^2]|_{(x,y)=(a,b)} \quad \textcircled{3}$$

• Case 1: $H(a,b) = (f_{xx}f_{yy} - f_{xy}^2)(a,b) > 0$.

By ③, we have $\forall (h,k) \neq (0,0)$, $f_{xx}(a,b)Q(0) > 0$.

1.1: $f_{xx}(a,b) > 0 \Rightarrow Q(0) > 0 \Rightarrow f(a,b)$ is a local min.

1.2: $f_{xx}(a,b) < 0 \Rightarrow Q(0) < 0 \Rightarrow f(a,b)$ is a local max.

Note that $f_{xx}(a,b)$ cannot be 0 in this case.

• Case 2: $H(a,b) = (f_{xx}f_{yy} - f_{xy}^2)(a,b) < 0$.

In this case, no matter how close (h,k) is to $(0,0)$,

Some (h,k) gives $Q(0) < 0$ while some (h,k) gives $Q(0) > 0$.

By continuity, the same statement can be made to $Q(c)$,

$$ax^2 + bx + c = a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) = a\left(x^2 + \frac{b}{a}x + \left(\frac{1}{2}\frac{b}{a}\right)^2 - \frac{1}{4}\left(\frac{b}{a}\right)^2\right) + c$$

$$= a\left(x + \frac{1}{2}\frac{b}{a}\right)^2 - \frac{a}{4}\frac{b^2}{a^2} + c = \frac{a}{4}\frac{b^2}{a^2} + c - \frac{b^2 - 4ac}{4a}$$

$$ax^2 + bx + c \neq 0$$

$$b^2 - 4ac \neq 0$$

$$4ac - b^2 < 0$$

$$b^2 - 4ac > 0$$

So $f(a,b)$ is neither a local max nor a local min.

$$t = \frac{h}{k}$$

(Details: By ②,

$$k > 0$$

$$Q(0) = \frac{1}{2}(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2)|_{(x,y)=(a,b)} = \frac{\frac{h}{k}}{2} \left(f_{xx}\left(\frac{h}{k}\right)^2 + 2f_{xy}\left(\frac{h}{k}\right) + f_{yy} \right)_{(x,y)=(a,b)}$$

Consider $P(t) = f_{xx}(a,b)t^2 + 2f_{xy}(a,b)t + f_{yy}(a,b)$

Case 1: $f_{xx}(a,b) \neq 0$

Since $(f_{xx}f_{yy} - f_{xy}^2)(a,b) < 0$, $P(t)$ has two real roots,

So $y = P(t)$ is a parabola crosses the t -axis twice, and is sometimes > 0 and sometimes < 0 . This means that for $(h,k) = (h_0, 1)$, $Q(0) = \frac{1}{2}P(h_0)$ is sometimes < 0 and sometimes > 0 . By scaling down $(h,k) = (h_0, 1)$, i.e., multiply $(h_0, 1)$ with a small constant, we see that $Q(0) = Q_{h,k}(0)$ takes different signs no matter how close (h,k) is to $(0,0)$.

Case 2: $f_{xx}(a,b) = 0$ $H < 0$

Then $f_{xy}(a,b) \neq 0$, so $y = P(t)$ is a line that takes both "+" and "-" values. The rest is similar to case 1.

$$Q(0) = f_{xx} \left(\quad \right)^2$$



By checking $f(x,y) = x^2$ and $g(x,y) = -x^2$, we see that

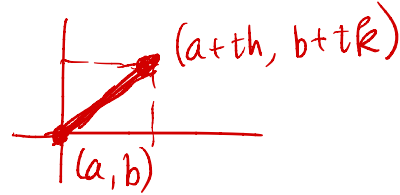
test is inconclusive when $H(a,b) = 0$.

$$F'(t) = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

Taylor's Theorem (Two-Variable Functions) ^h with cts second-order partials

We saw that if we are given f and $f(x, y)$, and let

$$F(t) := f(a+th, b+tk),$$



then

$$F'(t) = f_x h + f_y k = \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f$$

$$F''(t) = f_{xx} h^2 + 2f_{xy} hk + f_{yy} k^2 = \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f$$

More generally, if all n^{th} -order partials of f are cts on an open region containing (a, b) , then

$$F^{(n)}(t) = \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^n f, \quad \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^4$$

e.g. $F^{(4)}(t) = h^4 f_{xxxx} + 4h^3 k f_{xxx y} + 6h^2 k^2 f_{xx yy} + 4h k^3 f_{x y y y} + k^4 f_{yyyy}$

One variable Taylor's theorem suggests that $F(t) = F(0) + F'(0)t + \frac{1}{2}F''(0)t^2 + \dots + \frac{1}{n!}F^{(n)}(0)t^n + \frac{1}{(n+1)!}F^{(n+1)}(c)t^{n+1}$

$$F(t) = F(0) + F'(0)t + \frac{1}{2!}F^{(2)}(0)t^2 + \dots + \frac{1}{n!}F^{(n)}(0)t^n + \frac{1}{(n+1)!}F^{(n+1)}(c)t^{n+1}$$

for some $c \in (0, 1)$. This translates to the following two-variable version of Taylor's theorem: if $f(x, y)$ has continuous partials up to and including $(n+1)^{\text{th}}$ order in an open disk (or rectangular region) R containing (a, b) , then $\forall$ points in R ,

$$f(a+th, b+tk) = \left(\sum_{i=0}^n \frac{1}{i!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^i f \right) \Big|_{(a,b)} + \frac{1}{(n+1)!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^{n+1} f \Big|_{(a+tc, b+tk)}$$

$\mathcal{D}_{h,k}^n f$

for some $c \in (0,1)$.

Using Taylor expansion

EXAMPLE 14.9.1 Find a quadratic approximation to $f(x, y) = \sin x \sin y$ near the origin. How accurate is the approximation if $|x| \leq 0.1$ and $|y| \leq 0.1$?

Sol.

$$\begin{aligned} f(0,0) &= \sin x \sin y|_{(0,0)} = 0, & f_{xx}(0,0) &= -\sin x \sin y|_{(0,0)} = 0, \\ f_x(0,0) &= \cos x \sin y|_{(0,0)} = 0, & f_{xy}(0,0) &= \cos x \cos y|_{(0,0)} = 1, \\ f_y(0,0) &= \sin x \cos y|_{(0,0)} = 0, & f_{yy}(0,0) &= -\sin x \sin y|_{(0,0)} = 0. \end{aligned}$$

$$\begin{aligned} f(x,y) &= \sin x \sin y \\ f_x &= \cos x \sin y \\ f_y &= \sin x \cos y \\ f_{xx} &= -\sin x \sin y \\ f_{xy} &= \cos x \cos y \\ f_{yy} &= -\sin x \sin y \end{aligned}$$

$$f(x,y) \approx f(0,0) + (x f_x + y f_y)|_{(0,0)} + \frac{1}{2} (x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy})|_{(0,0)} \\ \approx xy$$

$$E(x,y) = \frac{1}{6} (x^3 f_{xxx} + 3x^2 y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy})|_{(0,0)}$$

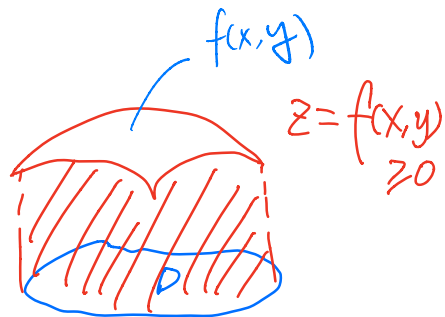
Computing all third-order partials shows that their absolute values are all ≤ 1 . $E \leq \frac{1}{6} (|x|^3 + 3|x|^2|y| + 3|xy|^2 + |y|^3)$

$$\text{Hence } |E(x,y)| \leq \frac{1}{6} (0.1^3 + 3(0.1)^3 + 3(0.1)^3 + (0.1)^3) \leq 0.00134.$$

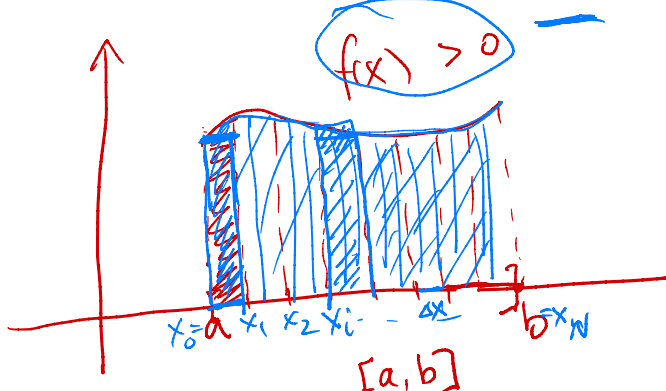
Double Integrals on Rectangles

What is the volume of this solid?

$$\iint_D f(x,y) dx dy$$

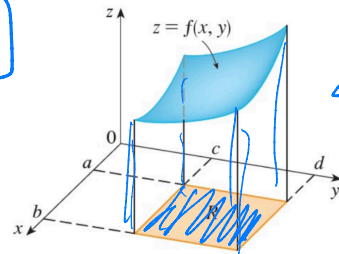


First look at case where D is a rectangle.



$$\Delta x = \frac{b-a}{N}$$

N large
 Δx small



$$\int_a^b f(x) dx = \text{Area under graph of } f(x)$$

$$\sum_{i=0}^{N-1} f(x_i) \Delta x \approx \text{Riemann sum}$$

Let $N \rightarrow \infty$ ($\Delta x \rightarrow 0$)

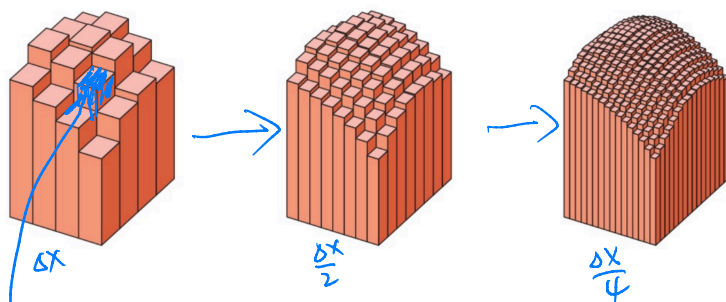
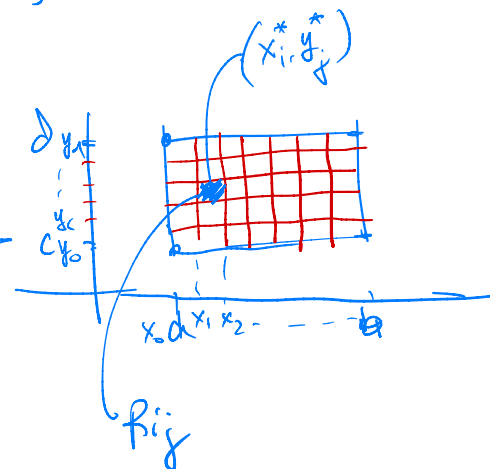
$$\sum_{i=0}^{N-1} f(x_i) \Delta x \rightarrow L = \int_a^b f(x) dx$$

as $N \rightarrow +\infty$

Let f be a two-variable nonnegative function defined on the rectangle

$$R := [a, b] \times [c, d] = \{(x, y) : a \leq x \leq b, c \leq y \leq d\}.$$

To approximate the volume of the solid lying between the xy -plane and the graph of f , we can use rectangular solids.



More precisely:

- Take a partition $\{x_0, x_1, \dots, x_m\}$ of $[a, b]$ and a partition $\{y_0, y_1, \dots, y_n\}$ of $[c, d]$; they together form a partition of R , and break R into mn rectangles R_{ij} .
- Let ΔA_{ij} be the area of R_{ij} , and let (x_{ij}^*, y_{ij}^*) be a point in R_{ij} .
- Then $f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}$ is the volume of the rectangular solid with base R_{ij} and height $f(x_{ij}^*, y_{ij}^*)$.
- The total volume is then approximated by the sum

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}.$$

The sum above is called a **Riemann sum** of f .

Definition

Let P be a partition of a rectangle R . The **norm** of P , denoted by $\|P\|$, is the largest width of the sub-rectangles in the partition P . For a function f defined on the rectangle R , if the limit of the Riemann sums

$$\sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}$$

exists as $\|P\| \rightarrow 0$, then f is said to be **integrable** (on R), and its limit is denoted by the **double integral**

$$\iint_R f(x, y) dA.$$

$$m \rightarrow +\infty$$

$$n \rightarrow +\infty$$

$$\|P\| \sim |\Delta x \cdot \Delta y|$$

$$\text{or } \Delta x \rightarrow 0$$

$$\Delta y \rightarrow 0$$

$$\text{or } \Delta A_{ij} \rightarrow 0$$

Remark

- ▶ Continuous functions are integrable; that is, if f is continuous on R , then

$$\iint_R f(x, y) dA$$

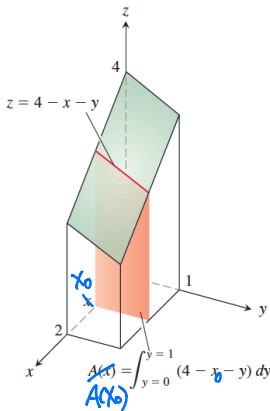
exists. In fact, if f is bounded on R , and is only discontinuous at finitely many points or on a finite union of smooth curves, then f is also integrable.

- ▶ If f is a nonnegative function defined on R , then the **volume** of the solid lying between the graph of f and the xy -plane is defined by

$$V := \iint_R f(x, y) dA.$$

Q: How to compute $\iint_R f(x, y) dA$?

e.g. Find the volume under the plane $z = 4 - x - y$ and over $R: 0 \leq x \leq 2, 0 \leq y \leq 1$ in the xy -plane



Approach 1: • Fix $x_0 \in [0, 2]$

- Consider thin "plate" on the left, whose area is $A(x_0)$

$$A(x_0) = \int_0^1 (4 - x_0 - y) dy = (4 - x_0) - \frac{1}{2}y^2 \Big|_0^1 = 4 - x_0 - \frac{1}{2}$$

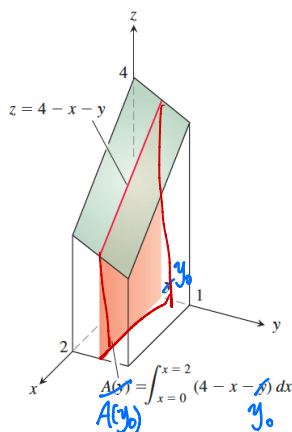
$$V = \int_0^2 A(x) dx = \int_0^2 \int_0^1 (4 - x - y) dy dx = \int_0^2 (4 - x - \frac{1}{2}) dx$$

One variable integrals, a few times $\rightarrow$ iterated integral

$$V = \int_0^2 \left((4y - xy - \frac{1}{2}y^2) \Big|_{y=0}^1 \right) dx = \int_0^2 (4 - x - \frac{1}{2}) dx$$

$$= \left(\frac{7}{2}x - \frac{1}{2}x^2 \right) \Big|_{x=0}^2 = 7 - 2 = 5.$$

Approach 2:



• Fix $y_0 \in [0, 1]$.

• $A(y_0) = \int_0^2 (4 - x - y_0) dx$

• $V = \int_0^1 A(y) dy = \int_0^1 \int_0^2 (4 - x - y) dx dy$

$= \int_0^1 \left[4x - \frac{1}{2}x^2 - yx \right]_{x=0}^2 dy$

$= \int_0^1 (8 - 2 - 2y) dy = (6y - y^2) \Big|_{y=0}^1 = 5$

Fubini's theorem states that the pattern above persists as long as the integrand is continuous on R .

Theorem (Fubini's Theorem) (for rectangles)

If f is continuous on $R := [a, b] \times [c, d]$, then

iterated integrals

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx.$$

Consequences: if f is cts on R :

- Double integrals can be computed using iterated integrals.
- Order does not matter.

Example

Evaluate $\iint_R y \sin(xy) dA$, where $R = [1, 2] \times [0, \pi]$.

$$\begin{aligned}
 &= \int_0^\pi \int_1^2 y \sin(xy) dy dx = \int_0^\pi \int_1^2 y \sin(xy) dx dy \\
 &= \int_0^\pi y \left(\int_1^2 \sin(xy) dx \right) dy \\
 &= \int_0^\pi y \left(-\cos(xy) \frac{1}{y} \right) \Big|_1^2 dy \\
 &= \int_0^\pi y \cdot \frac{1}{y} (-\cos(2y) + \cos(y)) dy \\
 &= \int_0^\pi (-\cos(2y) + \cos(y)) dy \\
 &= \dots
 \end{aligned}$$

Double Integrals on General Bounded Regions

Usually the case, but not essential $\leftarrow$ Essential

Let D be a closed and bounded region in $\mathbb{R}^2$, and suppose that f is a function defined on D . Let R be a rectangle enclosing D , and define a new function F on R by

$$F(x, y) := \begin{cases} f(x, y), & \text{if } (x, y) \in D; \\ 0, & \text{if } (x, y) \in R \setminus D. \end{cases}$$

We define the **double integral of f over D** by

$$\iint_D f(x, y) dA := \iint_R F(x, y) dA.$$

Again, if f is nonnegative and continuous on D , then we define the **volume** of the solid lying between the xy -plane and the graph of f to be $\iint_D f(x, y) dA$.

